

SIMATSENGINEERING

ASSESSMENTTOOL1

COURSENAME:ArtificialIntelligence

COURSECODE:CSA17

COURSEOUTCOMECOVERED

CO3:Analyzeknowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

AssessmentToolWeightage:50%

TimeDuration:1Hour

QUESTIONS: 4

TotalMarks:40

Annexure A

CASE STUDY

CodeofConduct:

I M.VAISHNAV(192524251) certifythat this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understandthatanyviolationofacademicintegrityruleswillresultindisciplinaryaction.

Signatureofthestudent:

M.VAISHNAV(192524251)

Annexure A

CASE STUDY

Case Study 1—Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever=True, Cough=True, Breathlessness=True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2—Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x, y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{Car A}), \text{Behind}(\text{Car A}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed(Car A)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3—Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) $P1: \text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) $P2: \text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) $P3: \neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences $P1, P2, P3$ and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4—Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell } [1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell } [1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at } [x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fail to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

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CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

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(a) Propositional Logic Representation

Symbols:

F = Patient has Fever
C = Patient has Cough
R = Patient has Rash
B = Patient has Breathlessness
FL = Patient has Flu
M = Patient has Measles
P = Patient has Pneumonia

Rules:

R1: $F \wedge C \rightarrow FL$
R2: $F \wedge R \rightarrow M$
R3: $C \wedge B \rightarrow P$

Facts:

F = True
C = True
B = True

Knowledge Base:

1. $F \wedge C \rightarrow FL$
2. $F \wedge R \rightarrow M$
3. $C \wedge B \rightarrow P$
4. F
5. C
6. B

(b) Forward Chaining

Initial Facts:

Fever = True
Cough = True
Breathlessness = True

Step 1:

Rule R1:
 $F \wedge C \rightarrow FL$

Since Fever=True and Cough=True

Infer:

FL = True

Diagnosis: Flu

Step 2:

Rule R3:
 $C \wedge B \rightarrow P$

Since Cough=True and Breathlessness=True

Infer:

$P = \text{True}$

Diagnosis: Pneumonia

Step 3:

Rule R2:
 $F \wedge R \rightarrow M$

Rash is not available.

Therefore Measles cannot be inferred.

Final Diagnoses:

1. Flu
2. Pneumonia

(c) Backward Chaining

Goal:

Prove Pneumonia

Goal:
 P

Using Rule:

$C \wedge B \rightarrow P$

Sub Goals:

$C ?$
 $B ?$

Facts Available:

$C = \text{True}$
 $B = \text{True}$

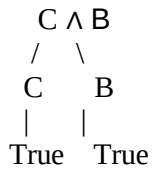
Both subgoals satisfied.

Therefore:

$P = \text{True}$

Goal Reduction Tree

P
|



Conclusion:

Patient A has Pneumonia.

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CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

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Knowledge Base

Rule 1:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x)$
 $\rightarrow \text{ClearPath}(x)$

Rule 2:

$\text{ClearPath}(x)$
 $\rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

Rule 3:

$\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y)$
 $\rightarrow \text{Proceed}(x)$

Facts:

$\text{Vehicle}(\text{Ambulance})$
 $\text{EmergencyType}(\text{Ambulance})$
 $\text{Vehicle}(\text{CarA})$
 $\text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Unification

Rule 1:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x)$
 $\rightarrow \text{ClearPath}(x)$

Facts:

$\text{Vehicle}(\text{Ambulance})$
 $\text{EmergencyType}(\text{Ambulance})$

Substitution:

$\theta = \{x/\text{Ambulance}\}$

After applying θ :

$\text{Vehicle}(\text{Ambulance})$
 $\wedge \text{EmergencyType}(\text{Ambulance})$

Therefore:

$\text{ClearPath}(\text{Ambulance})$

(b) Resolution Proof

Goal:

Proceed(CarA)

Convert to CNF

Rule 1:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x)$
 $\vee \text{ClearPath}(x)$

Rule 2:

$\neg \text{ClearPath}(x)$
 $\vee \text{GreenSignal}(x)$

Rule 3:

$\neg \text{Vehicle}(x)$
 $\vee \neg \text{Behind}(x,y)$
 $\vee \neg \text{GreenSignal}(y)$
 $\vee \text{Proceed}(x)$

Facts:

Vehicle(Ambulance)
EmergencyType(Ambulance)
Vehicle(CarA)
Behind(CarA,Ambulance)

Resolution Steps

Vehicle(Ambulance)
EmergencyType(Ambulance)

Infer:

ClearPath(Ambulance)

ClearPath(Ambulance)

Infer:

GreenSignal(Ambulance)

Vehicle(CarA)
Behind(CarA,Ambulance)
GreenSignal(Ambulance)

Apply Rule 3

Infer:

Proceed(CarA)

Hence Proved.

(c) Analysis

Current KB can manage only one emergency vehicle.

For two emergency vehicles, additional rules are needed.

Examples:

EmergencyPriority(Ambulance1,High)

EmergencyPriority(FireTruck,Medium)

EmergencyPriority(x,High)
 $\wedge$ EmergencyPriority(y,Medium)

$\rightarrow$ Serve(x) before Serve(y)

Additional facts:

Vehicle(Ambulance2)

EmergencyType(Ambulance2)

These rules help resolve conflicts and traffic priorities.

CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

Given

P1:
SoilDry $\rightarrow$ IrrigationNeeded

P2:
IrrigationNeeded $\wedge$ CropWheat
 $\rightarrow$ ApplyDripMethod

P3:
 $\neg$ ApplyDripMethod $\wedge$ CropWheat
 $\rightarrow$ CropAtRisk

Facts:

SoilDry
CropWheat

(a) CNF Conversion

P1

SoilDry $\rightarrow$ IrrigationNeeded

CNF:

$\neg$ SoilDry $\vee$ IrrigationNeeded

P2

$\text{IrrigationNeeded} \wedge \text{CropWheat}$
 $\rightarrow \text{ApplyDripMethod}$

CNF:

$\neg \text{IrrigationNeeded}$
 $\vee \neg \text{CropWheat}$
 $\vee \text{ApplyDripMethod}$

P3

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat}$
 $\rightarrow \text{CropAtRisk}$

CNF:

ApplyDripMethod
 $\vee \neg \text{CropWheat}$
 $\vee \text{CropAtRisk}$

Facts

SoilDry

CropWheat

CNF Clauses

1. $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$
 2. $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$
 3. $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$
 4. SoilDry
 5. CropWheat
-

(b) Resolution Refutation

Goal:

ApplyDripMethod

Negated Goal:

$\neg \text{ApplyDripMethod}$

Add to KB.

Step 1

Resolve

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

with

SoilDry

Result:

IrrigationNeeded

Step 2

Resolve

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

with

IrrigationNeeded

Result:

$\neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Step 3

Resolve

$\neg \text{CropWheat} \vee \text{ApplyDripMethod}$

with

CropWheat

Result:

ApplyDripMethod

Step 4

Resolve

ApplyDripMethod

with

$\neg \text{ApplyDripMethod}$

Result:

□

Empty Clause obtained.

Therefore:

ApplyDripMethod is proved.

Resolution Tree

SoilDry
|
IrrigationNeeded
|
CropWheat
|
ApplyDripMethod
|
¬ApplyDripMethod
|
□

(c) Removal of SoilDry

Remaining Facts:

CropWheat

Without SoilDry

Cannot derive IrrigationNeeded.

Without IrrigationNeeded

Cannot derive ApplyDripMethod.

Without proving ApplyDripMethod

CropAtRisk cannot be concluded.

Conclusion:

CropAtRisk does not logically follow.

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CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

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Percepts

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Rules

1. Stench[1,2]
→ Wumpus adjacent to [1,2]

2. Breeze[1,1]

→ Pit adjacent to [1,1]

3. Glitter[x,y]

→ Gold at [x,y]

4. $\neg$ Wumpus[x,y]

$\wedge \neg$ Pit[x,y]

→ Safe[x,y]

(a) Belief State using Modus Ponens

Fact:

Stench[1,2]

Infer:

Wumpus adjacent to [1,2]

Fact:

Breeze[1,1]

Infer:

Pit adjacent to [1,1]

Fact:

Glitter[2,2]

Infer:

Gold at [2,2]

Derived Facts

1. Wumpus adjacent to [1,2]

2. Pit adjacent to [1,1]

3. Gold at [2,2]

(b) Forward Chaining

From Stench[1,2]

Possible Wumpus Locations:

[1,1]

[1,3]

[2,2]

From Breeze[1,1]

Possible Pit Locations:

[1,2]

[2,1]

Conclusion

Possible Wumpus:

{[1,1],[1,3],[2,2]}

Possible Pit:

{[1,2],[2,1]}

(c) Path Safety Analysis

Path:

[1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

Cell [1,2]

Possible Pit Location

Therefore Unsafe

Cell [2,2]

Contains Gold

But may also be adjacent to Wumpus

Therefore uncertain

Final Conclusion

The path

[1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

cannot be guaranteed safe because [1,2] may contain a Pit and [2,2] may be dangerous due to Wumpus presence.

Hence the agent should not move directly along this path without obtaining additional information.

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FINAL CONCLUSION

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Case Study 1:

Diagnoses = Flu and Pneumonia

Case Study 2:

Proceed(CarA) proved using Resolution

Case Study 3:

ApplyDripMethod proved using Resolution Refutation

Case Study 4:

Gold located at [2,2] but path is not guaranteed safe

Therefore all tasks have been completed using Propositional Logic, FOL, Forward Chaining, Backward Chaining, Resolution, CNF Conversion, Unification and Logical Reasoning.